

WORLD CUP 2026

An interpretable expected-goals study

An inspectable shot-quality model, a local visual analytics interface, and a documented audit of the data behind both.

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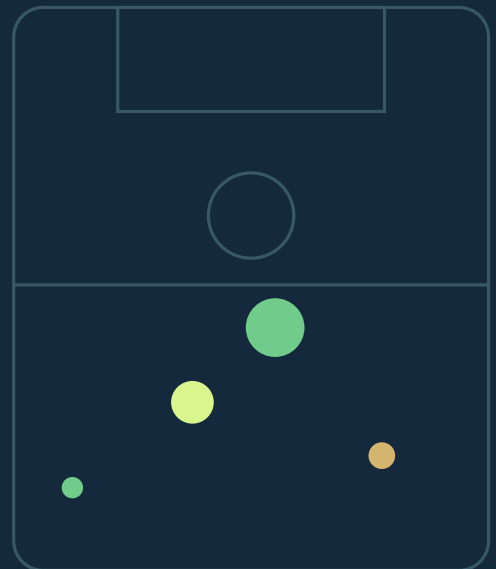
MATCHES

2,511

MODELLED SHOTS

0.765

TEST ROC AUC



THE RESEARCH QUESTION

How much of a football shot's probability of scoring can be explained by its location and observable event context?

Prepared for MSc Sport Data Science and AI

University of Birmingham | Technical portfolio supplement

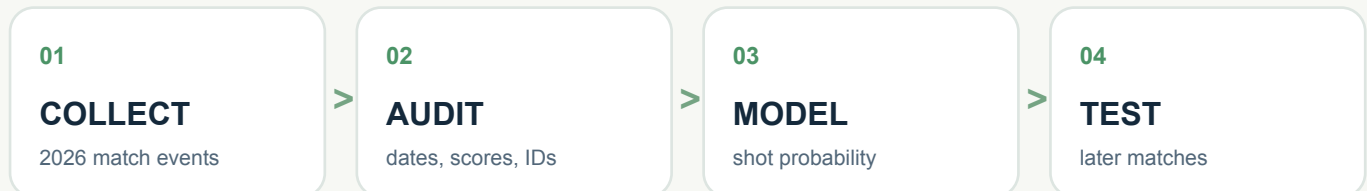
PUBLIC CC0 DEMO / <https://laojuzi123.github.io/touchline-world-cup-2026-cc0-demo/>

RESEARCH PREVIEW / EVENT-LEVEL DATA NOT INCLUDED

01 / STUDY DESIGN

A model built to be questioned

The unit of analysis is one non-penalty shot. A simple model makes the data preparation and evaluation choices inspectable.



TEMPORAL HOLDOUT

72 group-stage matches

1,738 shots / 195 goals

32 knockout matches

773 shots / 83 goals

FEATURES KNOWN AT SHOT TIME

GEOMETRY Distance, squared distance and opening angle to goal

TECHNIQUE Header indicator

SITUATION Set piece, direct free kick and fast break indicators

BUILD-UP Whether the shot was assisted

LEAKAGE CONTROL

Shot outcome, blocked status, post-shot location and the source's xG are excluded as predictors. Group-stage display values use five-fold match-grouped out-of-fold predictions; the knockout set stays untouched until evaluation.

02 / EVALUATION

Evidence, not just a score

All figures below are from the 32-match knockout holdout. Lower Brier score and log loss indicate better probabilistic predictions.

BRIER SCORE

0.0849

baseline 0.0959

LOG LOSS

0.2939

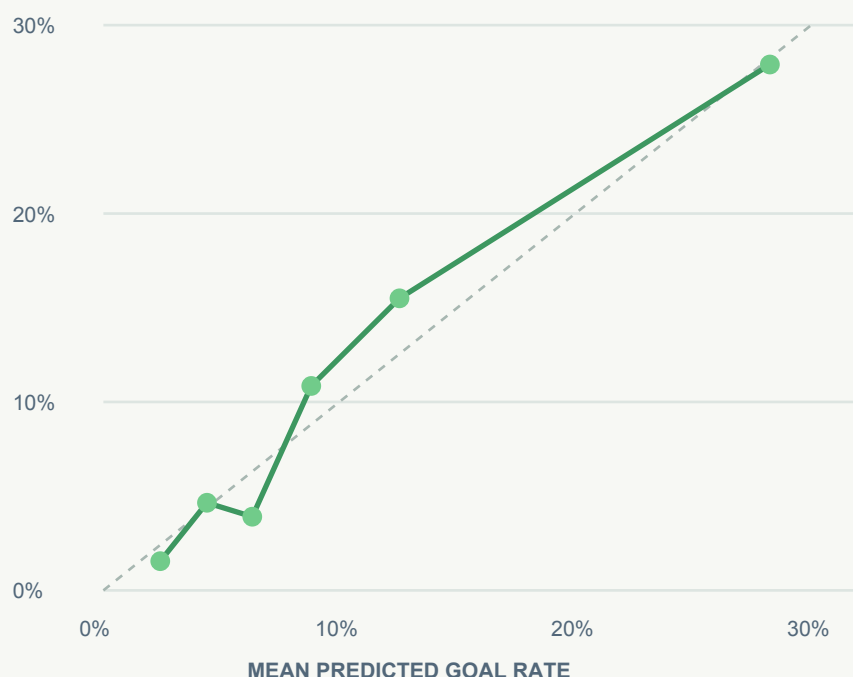
baseline 0.3411

ROC AUC

0.7649

baseline 0.5000

CALIBRATION BY PREDICTED-RISK BIN



HOW TO READ THIS

Points near the diagonal are better calibrated. The six bins each contain about 129 knockout shots.

WHAT IMPROVED

Brier score improved by 11.5% relative to the constant-rate baseline.

AUC = 0.765

INTERPRETATION

The holdout result suggests that basic shot geometry and context carry predictive information in this tournament. One competition, 773 test shots and 83 test goals are insufficient to establish general performance across leagues, seasons or providers.

03 / RESPONSIBLE PRACTICE

The data audit is part of the work

A credible sports analytics project records what was checked, what was corrected, and what cannot be established from public sources.

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fixtures and final scores matched an independent CC0 results table

44

source dates were one day later than venue-local match dates

5

extra-time matches needed final scores reconstructed from goal events

308

non-shootout goals, matching FIFA's tournament total

LIMITS OF VERIFICATION

Match-level checks cannot independently authenticate every shot coordinate or event qualifier. The event repository attributes its source to WhoScored but carries no explicit data reuse licence. This PDF omits event rows and shot locations. A separate CC0 match-results explorer is included as the shareable interactive demo.

WHY THIS FITS BIRMINGHAM**Programming for Data Science**

Reproducible Python ETL and validation

Sports Data Management & Visualisation

Data audit and interactive dashboard

Applied Match Analytics & Predictive Modelling

xG model and temporal holdout

SOURCE NOTESEvent source: github.com/nlbair/wc2026-events (commit 4674e5c; reuse licence unconfirmed)Match reference: github.com/openfootball/worldcup.json (commit 516d382; CC0-1.0)Official checks: fifa.com World Cup 2026 results; inside.fifa.com 2026 on-pitch totals